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Magnetic Resonance Imaging (MRI) of the Extremities

Clinical Policy Bulletins | Medical Clinical Policy Bulletins

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Policy History

Last Review 03/31/2026
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Review History

Definitions

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Policy

Scope of Policy

This Clinical Policy Bulletin addresses magnetic resonance imaging (MRI) of the extremities.

I. Medical Necessity

Aetna considers the following interventions medically necessary:

A. Magnetic resonance imaging (MRI) studies of the knee when *any* of the following criteria is met:

1. Detection, staging, and post-treatment evaluation of tumor of the knee; *or*
2. Persistent knee pain/swelling and/or instability (giving way) when:
   a. Not associated with an injury and not responding to at least 3 weeks of conservative therapy; *or*
   b. Secondary to an injury and not responding to conservative therapy when multi-view x-rays have ruled out a fracture or loose body in the knee and the clinical picture remains uncertain. **Note:** Conservative therapy consists of a combination of rest, ice, compression, elevation, non-steroidal anti-inflammatory drugs (NSAIDs), crutches, and range of motion (ROM) exercises; *or*
3. Persistent true locking of the knee indicative of a torn meniscus or loose body. (True locking is defined as more than a momentary locking of the joint with the knee in a flexed position, as compared to the sensation of momentary “catching” that many individuals experience in extension.); *or*
4. Suspected bone infection (i.e., osteomyelitis); *or*

5. Suspected osteochondritis dissecans or suspected osteonecrosis, if the clinical picture, including x-rays, is not confirmatory;

B. MRI for the diagnosis of CLOVES syndrome and for the diagnosis of osteomyelitis in the foot;
C. MRI for Morton neuroma for pre-operative planning when symptomatic neuroma has been identified by plain X-ray and non-surgical treatments (e.g., metatarsal support, padded shoe insert, and steroid/local anesthetic injections) have failed;
D. Whole-body MRI for screening of malignancy in persons with Li-Fraumeni syndrome;
E. MRI-lymphangiography for the evaluation of peripheral lymphedema when lymphedema is a suspected cause of peripheral edema and initial noninvasive studies, such as ultrasound, are negative.

II. **Experimental, Investigational, or Unproven**

The following interventions are considered experimental, investigational, or unproven because the effectiveness of these approaches has not been established:

A. Knee MRI for all other indications, including *any* of the following circumstances:

- Fitting of implants for total knee arthroplasty; *or*
- For chronological age estimation; *or*
- If arthroscopy or ligament reconstruction is definitely planned and the MRI findings are unlikely to change the planned treatment; *or*
- If the clinical picture (i.e., history, physical examination, x-rays, etc.) is diagnostic with high degree of certainty of a torn meniscus, loose body, or osteochondritis dessicans; *or*
- To diagnose or evaluate rheumatoid arthritis or degenerative joint disease;

B. MRI of the extremities (e.g., hands, knees, feet, etc.) for the following indications:

- Assessment of perfusion in diabetic foot ulcer
- Diagnosis or monitoring arthritis
- Diagnosis or prognosis of spinal cord injury and whiplash associated disorder
- Diagnosis of chronic exertional compartment syndrome
- Diagnosis suspected upper extremity deep vein thrombosis
- Evaluation and/or monitoring of disease progression in facioscapulohumeral muscular dystrophy;

C. Functional MRI for evaluation of upper extremity impairment after stroke;
D. Quantitative MRI for evaluation of muscle microtrauma in the lower extremity.

CPT Codes / HCPCS Codes / ICD-10 Codes

Code	Code Description
Magnetic resonance of knee:	
CPT codes covered if selection criteria are met:	
73721 - 73723	Magnetic resonance (e.g., proton) imaging, any joint of lower extremity [not covered for fitting of implant for total knee arthroplasty] [Not covered for chronological age estimation]
Other CPT codes related to the CPB:	
27427 - 27429	Ligamentous reconstruction (augmentation), knee

Code	Code Description
27437 - 27447	Arthroplasty, knee
29870 - 29889	Arthroscopy of knee
ICD-10 codes covered if selection criteria are met:	
C49.20 - C49.22	Malignant neoplasm of connective tissue of lower limb, including hip
C76.50 - C76.52	Malignant neoplasm of lower limb
D21.20 - D21.22	Benign neoplasm of connective and other soft tissue of lower limb, including hip
D48.0	Neoplasm of uncertain behavior of bone and articular cartilage
M23.000 - M23.92	Internal derangement of knee
M25.261 - M25.269	Flail joint, knee
M25.361 - M25.369	Other instability, knee
M25.461 - M25.469	Effusion, knee
M25.561 - M25.569	Pain in knee
M86.361 - M86.369 M86.461 - M86.469 M86.561 - M86.569 M86.661 - M86.669	Chronic osteomyelitis of lower leg
M87.051 - M87.066 M87.151 - M87.166 M87.251 - M87.266 M87.351 - M87.366 M87.851 - M87.869	Osteonecrosis, lower leg
M93.261 - M93.269	Osteochondritis dissecans, knee
S83.200A - S83.289S	Tear of meniscus, current injury
ICD-10 codes not covered for indications listed in the CPB:	
M05.00 - M14.89	Rheumatoid arthritis and other inflammatory polyarthropathies
M17.0 - M17.9	Osteoarthritis, knee
M22.40 - M22.42	Chondromalacia of patella
M23.8x1 - M23.92	Other internal derangements of knee
Z46.89	Encounter for fitting and adjustment of other specified devices [orthopedic devices] [implant for TKA]
MRI of extremities:	
CPT codes covered if selection criteria are met:	
73218 - 73223	Magnetic resonance (e.g., proton) imaging, upper extremity
73718 - 73723	Magnetic resonance (e.g., proton) imaging; lower extremity
CPT codes not covered for indications listed in the CPB:	
70554	Magnetic resonance imaging, brain, functional MRI; including test selection and administration of repetitive body part movement and/or visual stimulation, not requiring physician or psychologist administration
70555	requiring physician or psychologist administration of entire neurofunctional testing
Other CPT codes related to the CPB:	
64450	Injection(s), anesthetic agent(s) and/or steroid; other peripheral nerve or branch
Other HCPCS codes related to this CPB:	
L3030	Foot, insert, removable, formed to patient foot, each
L3050	Foot, arch support, removable, premolded, metatarsal, each
ICD-10 codes covered if selection criteria are met:	
G57.60 - G57.63	Lesion of plantar nerve
M86.8x7	Other osteomyelitis, ankle and foot

Code	Code Description
Q87.3	Congenital malformation syndromes involving early overgrowth [CLOVES syndrome]
ICD-10 codes not covered for indications listed in the CPB:	
G71.00 - G71.09	Muscular dystrophy [facioscapulohumeral muscular dystrophy]
I69.831 - I69.839	Monoplegia of upper limb following other cerebrovascular disease
I69.841 - I69.849	Monoplegia of lower limb following other cerebrovascular disease
I69.851 - I69.859	Hemiplegia and hemiparesis following other cerebrovascular disease
I69.861 - I69.869	Other paralytic syndrome following other cerebrovascular disease
M05.00 - M14.89	Rheumatoid arthritis and other inflammatory polyarthropathies
M62.10	Other rupture of muscle (nontraumatic), unspecified site
M62.151 - M62.159	Other rupture of muscle (nontraumatic), thigh
M62.161 - M62.169	Other rupture of muscle (nontraumatic), lower leg
M62.171 - M62.179	Other rupture of muscle (nontraumatic), ankle and foot
M62.89	Other specified disorders of muscle
M79.A11 - M79.A29	Nontraumatic compartment syndrome of upper and lower extremities
S13.4xxA-S13.4xxS	Sprain of ligaments of cervical spine
S14.0xxA-S14.9xxS	Unspecified injury of cervical spinal cord
S24.0xxA-S24.9xxS	Unspecified injury of thoracic spinal cord
S34.01xA-S34.139S	Unspecified injury to lumbar and sacral spinal cord
S86.911A - S86.919S	Strain of unspecified muscle and tendon at lower leg level
T79.A11A - T79.A29S	Traumatic compartment syndrome of upper and lower extremities
Whole-body MRI:	
CPT codes covered if selection criteria are met:	
Whole-body MRI – no specific code:	
ICD-10 codes covered if selection criteria are met:	
Z12.0 - Z12.9	Encounter for screening for malignant neoplasm [screening of malignancy with Li-Fraumeni syndrome]
Z15.01	Genetic susceptibility to malignant neoplasm of breast [screening of malignancy with Li-Fraumeni syndrome]
MRI-Lymphangiography:	
CPT codes covered if selection criteria are met:	
MRI-Lymphangiography –no specific code	
ICD-10 codes covered if selection criteria are met:	
I89.0	Lymphedema, not elsewhere classified

Background

Magnetic resonance imaging (MRI) has become the premier orthopedic diagnostic tool used in detecting meniscal and anterior cruciate ligament (ACL) tears and has virtually replaced both arthrography and arthroscopy as the diagnostic test of choice.

According to established guidelines from the American College of Rheumatology (2002), disease progression in rheumatoid arthritis (RA) should be followed using standard X-rays of the extremities. There is no adequate evidence from prospective clinical studies that clinical outcomes are improved by using MRI over standard X-rays for this indication. Although several studies have shown that MRI can detect early osseous changes, prospective clinical studies are needed to determine how well these early changes can predict development of clinically significant disease, and to determine whether clinical outcomes are improved by initiating therapy in persons with normal X-rays based on MRI findings. McQueen et al (2001) found that only 25% of changes detected by MRI progressed to X-ray erosions. These results raise questions about the nature and pathophysiologic basis of the osseous changes detected by MRI, whether one can predict which of these osseous changes will progress to X-ray erosions, and about the nature of the changes detected by MRI that do not progress to X-ray erosions.

In addition, prospective clinical studies are necessary to determine whether clinical outcomes are improved by using MRI over standard X-rays to monitor disease progression in persons with RA. Goldbach-Mansky et al (2003) of the National Institute of Arthritis and Musculoskeletal Diseases concluded that “[c]areful validation of MRI findings and the evaluation of MRI as a tool to follow the effect of therapy remain to be performed before MRI may be used as a clinical tool to follow therapy or as a surrogate for evaluating osseous changes over time.” Boutry et al (2005) evaluated prospectively the use of MRI for differentiating true RA from systemic lupus erythematosus (SLE) or primary Sjogren syndrome in patients who have inflammatory polyarthralgia of the hands but no radiographic evidence of RA. They concluded that it may be impossible to distinguish between patients with early RA and those without RA (namely those with SLE or primary Sjogren syndrome) by means of MRI. Graham et al (2005) reported that determining synovial volume in the hand and wrist in patients (n = 10) with juvenile rheumatoid arthritis by MRI is feasible and correlates with total hand swelling assessed on physical examination. However, these investigators stated that inconsistent or poor correlation with other clinical variables and the clinical definition of improvement requires further study.

Extremity MRI is not considered medically necessary to monitor the progression of arthritis. The American College of Rheumatology (ACR) Guidelines for the Management of Rheumatoid Arthritis (2002) lists X-rays as the appropriate method of monitoring disease progression. MRI of the arms and legs may be appropriate for the evaluation of masses, localized infections, non-healing fractures of long bones, and in certain cases, preoperative planning. Furthermore, the report on extremity MRI in RA by the ACR Extremity MRI Task Force (2006) stated that most of the literature assessing the utility of peripheral joint MRI has used high-field, not low-field extremity MRI; therefore, actual sensitivity, specificity, and predictive value of the low-field scanners available for the practicing rheumatologists are not known. The report also noted that the marginal benefit of low-field extremity MRI above and beyond standard measures of disease activity and severity (e.g., medical history, physical examination, selective laboratory testing, and radiography of the hands and wrists) has not been rigorously examined in studies published to date. In summary, the benefits of low-field strength extremity MRI for the diagnosis and management of RA are still being elucidated.

More recently, guidance on the management of rheumatoid arthritis from the National Institute for Health and Clinical Excellence (NICE, 2009) recommended X-ray the hands and feet early in the course of the disease in people with persistent synovitis in these joints. The Scottish Intercollegiate Guidelines Network (SIGN, 2011) concluded that "[t]he evidence for additional imaging at diagnosis to assess disease activity in early RA [rheumatoid arthritis] is limited and methodologically poor. The evidence suggests that power Doppler ultrasound may be useful in assessing disease activity and may have predictive value on radiological outcome." SIGN (2010) guidelines on psoriatic arthritis make no recommendation for magnetic resonance imaging of extremities.

There has been concern regarding the overuse of MRI. MRI has come to be perceived by many doctors and patients as the initial and the sine qua non diagnostic tool prior to surgical treatment. However, MRI should not be used as a routine screening tool in all knee injuries. Its use should be reserved for clinical situations in which the diagnosis remains in doubt. MRI does not replace a thorough history and physical examination and traditional multi-view x-rays as primary diagnostic tools. In a randomized controlled study (n = 87), Nikken and colleagues (2005) concluded that short MRI examination with a low-field-strength MRI system after radiography in initial evaluation of patients with acute wrist trauma has additional value in prediction of treatment need; however, it does not have value in identification of patients who can be discharged without further follow-up. In a randomized controlled study (n = 109), Oei and associates (2005) concluded that implementation of a dedicated extremity MRI examination, in addition to or instead of radiography, performed in patients with traumatic knee injury improves prediction of the need for additional treatment but does not significantly aid in identification of patients who can be discharged without further follow-up. They stated that value of a short MRI examination in the initial stage after knee trauma is limited.

Andrish (1996) stated that isolated meniscal injuries are rare in children under the age of 14, but the frequency increases thereafter. Meniscal tears in children are frequently associated with congenital meniscal abnormalities, while those in adolescents are often associated with ligamentous injuries of the knee. The combination of recurrent and often dramatic popping and intermittent episodes of locking has been termed the "snapping knee syndrome", and is almost invariably associated with a discoid meniscus. Although double-contrast arthrography has proved to be a reliable diagnostic technique, MRI is now the modality of choice. In this regard, Connolly et al (1996) had described the MRI appearance and associated abnormalities of discoid menisci in children. They noted that discoid meniscus commonly occurs bilaterally. High intra-meniscal signal is found, especially in symptomatic patients. The size criteria for diagnosing this condition in children are similar to those for adults.

In a prospective study, McNally et al (2002) examined if MRI of the acutely locked knee can alter surgical decision-making. The study group comprised patients with a clinical diagnosis of knee locking requiring arthroscopy. The decision to perform arthroscopy was made by an experienced consultant orthopedic surgeon specializing in trauma and recorded in the patient's notes prior to MRI. Pre-operative MRI was carried out using a 1.5 T system. The management was altered from surgical to conservative treatment in 20 (48%) patients on the basis of the MR findings. Arthroscopy was limited to patients with an MR diagnosis of a mechanical block, usually a displaced meniscal tear or loose body. Both patient groups were followed clinically until symptoms resolved. A total of 42 patients were entered into the study. MRI identified a mechanical cause for locking in 22 patients (21 avulsion meniscal tears and 1 loose body). All were confirmed at arthroscopy. Twenty patients were changed from operative to non-operative treatment on the basis of the MRI findings. One patient in this group required a delayed arthroscopy for an impinging anterior cruciate ligament stump. The sensitivity, specificity, and accuracy of MRI in identifying patients who require arthroscopy was 96, 100, and 98%, respectively. These investigators concluded that MRI can successfully segregate patients with a clinical diagnosis of mechanical locking into those who have a true mechanical block and those who can be treated conservatively. They stated that MRI should precede arthroscopy in this clinical setting.

Schurmann et al (2007) stated that complex regional pain syndrome type I (CRPS I) is difficult to diagnose in post-traumatic patients. As CRPS I is a clinical diagnosis the characteristic symptoms have to be differentiated from normal post-traumatic states. These researchers compared several diagnostic procedures for diagnosing post-traumatic CRPS I. A total of 158 patients with distal radial fracture were included in this study. A detailed clinical examination was carried out 2, 8, and 16 weeks after trauma in conjunction with bilateral thermography, plain radiographs of the hand skeleton, three phase bone scans (TPBSs), and contrast-enhanced magnetic resonance imaging (MRI). All imaging procedures were assessed blinded. At the end of the observation period, 18 patients (11%) were clinically identified as having CRPS I and 13 patients (8%) revealed an incomplete clinical picture which were defined as CRPS borderline cases. The sensitivity of all diagnostic procedures used was poor and decreased between the first and the last examinations (thermography: 45% to 29%; TPBS: 19% to 14%; MRI: 43% to 13%; bilateral radiographs: 36%). In contrast, a high specificity was observed in the TPBS and MRI at the 8th and 16th-week examinations (TPBS: 96%, 100%; MRI: 78%, 98%) and for bilateral radiographs 8 weeks after trauma (94%). Thermography presented a fair specificity that improved from the 2nd to the 16th week (50% to 89%). The authors concluded that the poor sensitivity of all tested procedures combined with a reasonable specificity produced a low positive predictive value (17% to 60%) and a moderate negative predictive value (79% to 86%). These results suggested that those procedures can not be used as screening tests. Imaging methods are not able to reliably differentiate between normal post-traumatic changes and changes due to CRPS I. Clinical findings remain the gold standard for the diagnosis of CRPS I and the procedures described above may serve as additional tools to establish the diagnosis in doubtful cases.

Tsai and Beredjiklian (2008) noted that arthritis of the thumb joints is a common problem and remains a significant cause of morbidity in the adult population. Careful physical examination is critical in the evaluation of these individuals, given the large differential diagnosis of conditions affecting the thumb and the radial side of the wrist. Because treatment should be specifically directed at the area of pathology, adequate diagnosis is vital. Plain radiograph examination remains the diagnostic modality of choice in the evaluation of patients with degenerative conditions regarding the hand and wrist.

The American College of Occupational and Environmental Medicine's clinical guideline on "Hand, wrist, and forearm disorders not including carpal tunnel syndrome" (2011) does not recommend MRI for diagnosing tuft fractures as well as phalangeal and metacarpal fractures.

Aweid et al (2012) stated that although all intra-compartmental pressure (ICP) measurement, MRI, and near-infrared spectroscopy seem to be useful in confirming the diagnosis of chronic exertional compartment syndrome (CECS), no standard diagnostic procedure is currently universally accepted. These researchers reviewed systematically the relevant published evidence on diagnostic criteria commonly in use for CECS to address 3 main questions: (i) Is there a standard diagnostic method available? (ii) What ICP threshold criteria should be used for diagnosing CECS? and (iii) What are the criteria and options for surgical management? Finally, these investigators made statements on the strength of each diagnostic criterion of ICP based on a rigorous standardized process. The authors searched for studies that investigated ICP measurements in diagnosing CECS in the leg of human subjects, using PubMed, Score, PEDRO, Cochrane, Scopus, SportDiscus, Web of Knowledge, and Google Scholar. Initial searches were performed using the phrase, "chronic exertional compartment syndrome". The phrase "compartment syndrome" was then combined, using Boolean connectors ("OR" and "AND") with the words "diagnosis", "parameters", "levels", "localisation," or "measurement". Data extracted from each study included study design, number of subjects, number of controls, ICP instrument used, compartments measured, limb position during measurements, catheter position, exercise protocol, timing of measurements, mean resting compartment pressures, mean maximal compartment pressures, mean post-exercise compartment pressures, diagnostic criteria used, and whether a reference diagnostic standard was used. The quality of studies was assessed based on the approach used by the American Academy of Orthopaedic Surgeons in judging the quality of diagnostic studies, and recommendations were made regarding each ICP diagnostic criteria in the literature by taking into account the quality and quantity of the available studies proposing each criterion. A total of 32 studies were included in this review. The studies varied in the ICP measurement techniques used; the most commonly measured compartment was the anterior muscle compartment, and the exercise protocol varied between running, walking, and ankle plantarflexion and dorsiflexion exercises. Pre-exercise, mean values ranged from 7.4 to 50.8 mm Hg for CECS patients, and 5.7 to 12 mm Hg in controls; measurements during exercise showed mean pressure readings ranging from 42 to 150 mm Hg in patients and 28 to 141 mm Hg in controls. No overlap between subjects and controls in mean ICP measurements was found at the 1-min post-exercise timing interval only showing values ranging from 34 to 55.4 mm Hg and 9 to 19 mm Hg in CECS patients and controls, respectively. The quality of the studies was generally not high, and the researchers found the evidence for commonly used ICP criteria in diagnosing CECS to be weak. The authors concluded that studies in which an independent, blinded comparison is made with a valid reference standard among consecutive patients are yet to be undertaken. There should also be an agreed ICP test protocol for diagnosing CECS because the variability here contributes to the large differences in ICP measurements and hence diagnostic thresholds between studies. Current ICP pressure criteria for CECS diagnosis are therefore unreliable, and emphasis should remain on good history. However, clinicians may consider measurements taken at 1 min after exercise because mean levels at this timing interval only did not overlap between subjects and controls in the studies that were analyzed. Levels above the highest reported value for controls here (27.5 mm Hg) along with a good history, should be regarded as highly suggestive of CECS. The authors stated that it is evident that to achieve an objective recommendation for ICP threshold, there is a need to set up a multi-center study group to reach an agreed testing protocol and modify the preliminary recommendations they have made.

Krabben et al (2013) stated that MRI is increasingly used to measure inflammation in rheumatoid arthritis (RA) research, but the correlation to clinical assessment is unexplored. This study determined the association and concordance between inflammation of small joints measured with MRI and physical examination. A total of 179 patients with early arthritis underwent a 68 tender joint count and 66 swollen joint count and 1.5T MRI of MCP (2-5), wrist and MTP (1-5) joints at the most painful side. Two readers scored synovitis and bone marrow edema (BME) according to the OMERACT RA MRI scoring method and assessed tenosynovitis. The MRI data were first analyzed continuously and then dichotomized to analyze the concordance with inflammation at joint examination. A total of 1,790 joints of 179 patients were studied. Synovitis and tenosynovitis on MRI were independently associated with clinical swelling, in contrast to BME. In 86% of the swollen MCP joints and in 92% of the swollen wrist joints any inflammation on MRI was present. In 27% of the non-swollen MCP joints and in 66% of the non-swollen wrist joints any MRI inflammation was present. Vice versa, of all MCP, wrist and MTP joints with inflammation on MRI 64%, 61% and 77%, respectively, were not swollen. Bone marrow edema, also in case of severe lesions, occurred frequently in clinically non-swollen joints. Similar results were observed for joint tenderness. The authors concluded that inflammation on MRI is not only present in clinically swollen but also in non-swollen joints. In particular BME occurred in clinically non-inflamed joints. The relevance of subclinical inflammation for the disease course is a subject for further studies.

Furthermore, Axelsen and colleagues (2014) examined the ability of whole-body MRI (WBMRI) to visualize inflammation [synovitis, BME and enthesitis] and structural damage in patients with RA. The 3T WBMR images were acquired in a head-to-toe scan in 20 patients with RA and at least 1 swollen or tender joint. Short Tau Inversion Recovery and pre- and post-contrast T1-weighted images were evaluated for readability and the presence/absence of inflammation (synovitis, BME and enthesitis) and structural damage (erosions and fat infiltrations) in 76 peripheral joints, 30 entheseal sites and in the spine. The readability was greater than 70% for all individual joints, except for the most peripheral joints of the hands and feet. Synovitis was most frequent in the wrist, first tarsometatarsal, first CMC joints and glenohumeral joints (67 to 61%); BME in the wrist, CMC, acromioclavicular and glenohumeral joints (45 to 35%) and erosions in the wrist, MTP and CMC joints (19 to 16%). Enthesitis at greater than or equal to 1 site was registered in 16 patients. Bone marrow edema was frequently seen in the cervical (20%) but not the thoracic and lumbar spine, while fat infiltrations and erosions were rare. The intra-reader agreement was high (85 to 100%) for all pathologies. The agreement between WBMRI and clinical findings was low. The authors concluded that peripheral and axial inflammation and structural damage at joints and entheses was frequently identified by WBMRI, and more frequently than by clinical examination. They stated that WBMRI is a promising tool for evaluation of the total inflammatory load of inflammation (an MRI joint count) and structural damage in RA patients.

Management of Diabetic Foot Ulceration

Forsythe and Hinchliffe (2016) noted that evaluation of foot perfusion is a vital step in the management of patients with diabetic foot ulceration, in order to understand the risk of amputation and likelihood of wound healing. Underlying peripheral artery disease (PAD) is a common finding in patients with foot ulceration and is associated with poor outcomes. Evaluation of foot perfusion should therefore focus on identifying the presence of PAD and to subsequently estimate the effect this may have on wound healing. Assessment of perfusion can be difficult because of the often complex, diffuse and distal nature of PAD in patients with diabetes, as well as poor collateralization and heavy vascular calcification. Conventional methods of evaluating tissue perfusion in the peripheral circulation may be unreliable in patients with diabetes, thus, it may therefore be difficult to determine the extent to which poor perfusion contributes to foot ulceration. Anatomical data obtained on cross-sectional imaging is important but must be combined with measurements of tissue perfusion (such as transcutaneous oxygen tension) in order to understand the global and regional perfusion deficit present in a patient with diabetic foot ulceration. Ankle-brachial pressure index is routinely used to screen for PAD, but its use in patients with diabetes is limited in the presence of neuropathy and medial arterial calcification. Toe pressure index may be more useful because of the relative sparing of pedal arteries from medial calcification but may not always be possible in patients with ulceration. Fluorescence angiography is a non-invasive technique that can provide rapid quantitative information about regional tissue perfusion; capillaroscopy, iontophoresis and hyper-spectral imaging may also be useful in assessing physiological perfusion but are not widely available. The authors concluded that there may be a future role for specialized perfusion imaging of these patients, including MRI techniques, single-photon emission computed tomography (SPECT) and positron emission tomography (PET)-based molecular imaging; however, these novel techniques require further validation and are unlikely to become standard practice in the near future.

Evaluation and/or Monitoring of Disease Progression in Facioscapulohumeral Muscular Dystrophy

Andersen and associates (2017) noted that there is no effective treatment available for facioscapulohumeral muscular dystrophy type 1 (FSHD1), but emerging therapies are underway that call for a better understanding of natural history in this condition. In a prospective, longitudinal study, these researchers used quantitative MRI to evaluate yearly disease progression in patients with FSHD1. Ambulatory patients with confirmed diagnosis of FSHD1 (25/20 men/women, age range of 20 to 75 years, FSHD score: 0 to 12) were tested with 359-560-day interval between tests. Using the MRI Dixon technique, muscle fat replacement was evaluated in para-spinal, thigh, and calf muscles. Changes were compared with those in FSHD score, muscle strength (hand-held dynamometry), 6-minute-walk-distance (6MWD), 14-step-stair-test, and 5-time-sit-to-stand-test. Composite absolute fat fraction of all assessed muscles increased by 0.036 (confidence interval [CI]: 0.026 to 0.046, p < 0.001), with increases in all measured muscle groups. The clinical severity FSHD score worsened (10%, p < 0.05), muscle strength decreased over the hip (8%), neck (8%), and back (17%) (p < 0.05), but other strength measures, 6MWD, 5-times-sit-to-stand-test, and 14-step-stair-test were unchanged. Changes in muscle strength, FSHD score, and fat fraction did not correlate. The authors concluded that this first study to systemically monitor quantitative fat replacement longitudinally in FSHD1 showed that MRI provided an objective measure of disease progression, often before changes can be appreciated in strength and functional tests. The study indicated that quantitative MRI can be a helpful end-point in follow-up and therapeutic trials of patients with FSHD1. These preliminary findings need to be validated by well-designed studies.

In a single-center study, Mui and colleagues (2017) examined if quantitative muscle MRI should be added to the clinical trial toolbox for(FSHD by correlating it to clinical outcome measures in a large cohort of genetically and clinically well-characterized patients with FSHD comprising the entire clinical spectrum. Quantitative MRI scans of leg muscles of 140 patients with FSHD1 and FSHD2 were assessed for fatty infiltration and TIRM hyper-intensities and were correlated to multiple clinical outcome measures. The mean fat fraction of the total leg musculature correlated highly with the motor function measure, FSHD clinical score, Ricci score, and 6MWD (correlation coefficients -0.845, 0.835, 0.791, -0.701, respectively). Fat fraction per muscle group correlated well with corresponding muscle strength (correlation coefficients up to -0.82). The hamstring muscles, adductor muscles, rectus femoris, and gastrocnemius medialis were affected most frequently, also in early stage disease and in patients without leg muscle weakness. Muscle involvement was asymmetric in 20% of all muscle pairs and fatty infiltration within muscles showed a decrease from distal to proximal of 3.9%. TIRM hyper-intense areas, suggesting inflammation, were found in 3.5% of all muscles, with and without fatty infiltration. The authors concluded that this study showed a strong correlation of quantitative muscle MRI with different clinical outcome measures, especially with the motor function measure. Since quantitative muscle MRI also had good clinimetric properties, it is a promising biomarker representative of disease severity. For clinical trials, these researchers proposed to include quantitative muscle MRI and the motor function measure in the FSHD trial toolbox. Moreover, they stated that while the single-center, single-evaluator design of this study avoided interrater variability and increased the reliability of correlations between fat fractions and clinical measures; however, a limitation of this design was the decrease in the external validity. Furthermore, results regarding disease duration should be interpreted with caution since age at disease onset was collected retrospectively.

Diagnosis and/or Prognosis of Spinal Cord Injury and Whiplash Associated Disorder

McPherson and colleagues (2018) stated that diffusion-weighted MRI (DW-MRI) of skeletal muscle has the potential to be a sensitive diagnostic and/or prognostic tool in complex, enigmatic neuro-musculoskeletal conditions such as spinal cord injury (SCI) and whiplash associated disorder. However, the reliability and reproducibility of clinically accessible DW-MRI parameters in skeletal muscle remains incompletely characterized -- even in individuals without neuro-musculoskeletal injury -- and these parameters have yet to be characterized for many clinical populations. These researchers provided normative measures of the apparent diffusion coefficient (ADC) in healthy muscles of the lower limb; evaluate the rater-based reliability and short- and long-term reproducibility of the ADC in the same muscles; and quantify ADC of these muscles in individuals with motor incomplete SCI. A total of 20 individuals without neuro-musculoskeletal injury and 14 individuals with motor incomplete SCI participated in this study. These investigators acquired bilateral DW-MRI of the lower limb musculature in all participants at 3-T using a multi-shot echo-planar imaging sequence with b-values of 0, 100, 300 and 500 s/mm2 and diffusion-probing gradients applied in 3 orthogonal directions. Outcome measures included average ADC in the lateral and medial gastrocnemius, tibialis anterior, and soleus of individuals without neurological or musculoskeletal injury; intra- and inter-rater reliability, as well as short- and long-term reproducibility of the ADC; and estimation of average muscle ADC in individuals with SCI. Intra- and inter-rater reliability of the ADC averaged 0.89 and 0.79, respectively, across muscles. Least significant change, a measure of temporal reproducibility, was 4.50 and 11.98% for short (same day) and long (9-month) inter-scan intervals, respectively. Average ADC was significantly elevated across muscles in individuals with SCI compared to individuals without neurological or musculoskeletal injury (1.655 versus 1.615 mm2/s, respectively). The authors concluded that despite the promise of DW-MRI in skeletal muscle as an additional tool for clinical evaluation and basic scientific investigation, its utility has remained limited at least in part due to a lack of normative data and an incomplete assessment of its precision and reliability. These results demonstrated that its reproducibility is excellent over short time scales, reinforcing the available literature. They also extended these findings by providing insights into the stability of the approach over the longest inter-scan interval available to-date (9 months). Although performance was expectedly lower over this time scale, ADC changes as small as 12% were still discernable. Considering that ADC changes of approximately 24% have been reported in radiculopathy, approximately 35% following rotator cuff tear, and approximately 20% immediately following vigorous contraction of the same muscles included in this investigation, the authors stated that controlled experiments using DW-MRI of skeletal muscle are sufficiently reproducible for longitudinal analyses. With evolving scanner and sequence technologies, DW-MRI may prove to be a highly sensitive measure of physiological changes to peripheral muscles after neurological injury and in any number of common, yet enigmatic, neuro-musculoskeletal conditions (e.g., low back pain, whiplash). Such knowledge could help to improve clinical decision-making at a time when therapeutic intervention may have its largest effect. These researchers stated that the findings of this provided a foundation for future studies that track longitudinal changes in skeletal muscle ADC of the lower extremity and/or examine the mechanisms underlying ADC changes in cases of known or suspected pathology.

The authors stated that this study had several drawbacks. Unlike previous investigations, their ROI definition was performed by raters with minimal imaging experience yet an intimate knowledge of the relevant anatomical structures (Doctor of Physical Therapy students). Although this choice was motivated by a desire to estimate a potential lower bound on rater-based reliability, it also complicated the interpretation of these findings by making it harder to decouple the sources of experimental, physiological, and rater-induced variability. Another drawback was that these researchers used the same MRI scanner for all imaging sessions. As a result, the ADC values, as well as their reliability estimates, should be interpreted only in this context. In purest terms, these findings may over-estimate rater-based reliability and/or reproducibility and thus be much different if compared to results from multi-site investigations incorporating different community imaging centers. Finally, these investigators did not control leg temperature in the cohort of individuals with SCI. Thus, it was possible that subtle temperature differences existed between the SCI and control cohorts and/or between the more and less impaired legs of individuals with SCI. Reduced temperature in the extremities, as has been reported in SCI, could change the estimates of ADC independently of SCI-induced alterations of muscle physiology. It appeared unlikely that changes in temperature could explain the differences in ADC they found between individuals with and without SCI, however. Specifically, leg temperature in individuals with SCI was generally lower than that of individuals without SCI, suggesting that ADC should be lower in SCI than controls. However, the authors found a significantly elevated ADC across muscles in the SCI cohort compared to the control cohort, indicating that temperature was not the primary factor driving the observed ADC differences. These researchers also found no significant differences in ADC between legs within the SCI cohort for any muscle. This finding also suggested that the influence of leg temperature on these findings was likely minimal (although it was presumably a source of overall variability in the estimates of ADC). Nevertheless, future work investigating skeletal muscle ADC in individuals with SCI (or other neurological injuries) should systematically monitor and control extremity temperature to avoid undue experimental confounds.

Diagnosis of CLOVES syndrome

Alomari and colleagues (2011) stated CLOVES syndrome, characterized by Congenital Lipomatous Overgrowth, Vascular malformations, Epidermal nevi, and Skeletal anomalies, is a complex disorder of congenital lipomatous overgrowth, vascular malformations, epidermal nevi, and skeletal/scoliosis/spinal anomalies. The authors reported the occurrence of spinal-paraspinal fast-flow lesions within or adjacent to the truncal over-growth or a cutaneous birthmark in 6 patients with CLOVES syndrome. The available imaging studies included MRI in 6 patients, CT in 4, and spinal angiography in 5.

Boston Children’s Hospital CLOVES Syndrome Work Group’s clinical practice guidelines on “CLOVES syndrome” (2014) stated that “CLOVES patients may benefit from an MRI of the chest, abdomen, pelvis and lower extremities performed in the neonatal or early infantile period or at the time of initial presentation. This help define deeper components of the syndrome that may require from intervention in early childhood (e.g., lymphatic and venous malformations, gastrointestinal and genitourinary involvement); as well as characterize overgrowth and extension into the retroperitoneum, peritoneum, superior and posterior mediastinum, pelvis, pleural spaces and paraspinal muscles, tethered spinal cord, neural tube defects. If the study requires general anesthesia, the scan can be delayed until the risk of anesthesia is reduced (i.e., greater than 6 months of age). Optimal timing of imaging best balances the risks of anesthesia with the information to be gained”.

Panteliades and associates (2016) stated that CLOVES syndrome is a rare, newly described, and relatively unknown syndrome, related to somatic mutations of the PIK3CA gene. Clinical findings include adipose tissue over-growth, vascular malformations, epidermal nevi, scoliosis, and spinal deformities. These researchers addressed a characteristic phenotype case, highlighting peculiar cutaneous and radiological changes. A 2-year old child was referred to evaluate the increase in the soft parts in the lower back, left arm, and posterior portion of the lower limbs, which had been present since birth. The patient also presented with flat feet, epidermal nevi on the right arm, nodular plate with a cystic formation on the left arm, spacing between the 1st and 2nd toes, capillary malformation in the trunk, as well as epispadia and scoliosis of the thoracic-lumbar spine. Imaging examinations (CT of the abdomen and MRI of the abdomen and right upper limb) revealed confluent and heterogeneous lobes located in the subcutaneous adipose tissue of the abdomen, with a cystic aspect, suggestive of lymphangioma, an asymmetrical increase in the fat deposit on the retro-peritoneal posterior abdominal wall, as well as para-vertebral muscles, presence of scoliosis in the thoracic-lumbar spine, and complex vascular malformation in the left arm.

Mahajan and co-workers (2019) stated that CLOVES syndrome is a recently described sporadic syndrome from post-zygotic activating mutations in PIK3CA. These researchers presented the case of a 3-year old boy, born to non-consanguineous and healthy parents, had epidermal verrucous nevus, lower limb length discrepancy and bilateral genuvalgum, anterior abdominal wall lipomatous mass, central beaking of L2 and L3, and fibrous dysplasia of the left frontal bone. Ocular and dental abnormalities (ptosis, esotropia, delayed canine eruption, dental hypoplasia), ipsilateral asymmetrical deformity of skull, and large left cerebral hemisphere with mild ipsilateral ventriculomegaly were peculiar to him denoting an uncommon phenotype. The parents did not consent for MRI and genetic studies because of financial constraints. The authors noted that CLOVES syndrome has emerged as an uncommon yet distinct clinical entity with some phenotypic variations. Its diagnosis is usually from cutaneous, truncal, spinal, and foot anomalies in clinical and radio-imaging studies.

Diagnosis of Osteomyelitis in the Foot

Jones and colleagues (2012) noted that the clinical presentation of acute Charcot arthropathy in the diabetic population usually follows the Eichenholtz classification. These researchers discussed 3 usual cases of Charcot arthropathy presenting with rapid primary bone resorption in the absence of subluxation, dislocation and/or fracture. They carried out a review of the literature. To their knowledge, Charcot arthropathy has not been previously described as primary bone resorption. Three cases encountered at the authors' multi-disciplinary High Risk Foot Clinic (HRFC) presented with primary bony resorption without features of subluxation, dislocation and/or fracture. Aggressive primary bone resorption was initially thought due to infection; a diagnostic dilemma that delayed optimal treatment. Late bone resorption in typical Charcot is linked to un-regulated proinflammatory cytokines (IL-1β, IL-6 and TNFα) that lead to increased osteoclastic activity. The pathophysiology of osteolysis in aggressive primary bony resorption may relate to a disturbance in the balance between RANK-L and OPG. The authors concluded that primary resorption of bone without subluxation, dislocation and/or fracture could represent an active Charcot process. These investigators stated that prudent use of serial radiography and early MRI to look for the widespread bone and soft tissue edema is recommended.

Madan and Pai (2013) stated that MRI is becoming a popular investigation for a variety of foot problems. In chronic charcot neuroarthropathy (CN), MRI shows low signal intensity in subchondral bone on both T1 and T2 weighted images; this correlates with sclerosis on radiographs. In contrast, in osteomyelitis there is low signal intensity on T1-weighted and high signal intensity on T2-weighted images. However, differentiating acute Charcot foot from osteomyelitis remains a challenge. The very similar clinical presentation of these 2 entities causes confusion in the diagnosis and overlap of the MRI findings adds to this confusion. However, there are some clues that can help to clinch the diagnosis. On MRI, a Charcot foot will show localized juxta-articular edema, whereas in osteomyelitis the edema is more on one side of the joint and is not confined to the juxta-articular area. Clinically, osteomyelitis affects a single bone in the fore-foot or hind-foot, whereas CN affects many bones, commonly in the mid-foot.

Whole-Body MRI for Screening of Malignancy in Adults with Li-Fraumeni Syndrome

Huby and colleagues (2019) stated that Li-Fraumeni syndrome (LFS) is a rare inherited disease characterized by the early onset of multiple primary malignant tumors. Sarcomas account for more than 30% of all malignant tumors occurring at pediatric age. Furthermore, it was shown that the rates of 2nd cancer were higher in childhood cancer survivors. These investigators reported the case of a patient with LFS who was referred to these researchers with 3 synchronous skeletal tumors. This unique situation led to difficulties for the medical team regarding the diagnosis of malignancy and the surgical treatment to propose. The discovery of multiple lesions in the extension assessment underlined the usefulness of whole-body imaging for the follow-up of patients with germline TP53 mutations. Most recent guidelines now recommend annual whole-body MRI (WB-MRI) for screening for cancer patients carrying germline TP53.

Frebourg and associates (2020) noted that 50 years after the recognition of the LFS, the perception of cancers related to germline alterations of TP53 has drastically changed: germline TP53 alterations are often identified among children with cancers, in particular soft-tissue sarcomas, adrenocortical carcinomas, central nervous system (CNS) tumors, or among women with early breast cancers, without familial history. This justifies the expansion of the LFS concept to a wider cancer predisposition syndrome designated heritable TP53-related cancer (hTP53rc) syndrome. The interpretation of germline TP53 variants remains challenging and should integrate epidemiological, phenotypical, bioinformatics prediction, and functional data. The penetrance of germline disease-causing TP53 variants is variable, depending both on the type of variant (dominant-negative variants being associated with a higher cancer risk) and on modifying factors. These investigators stated that WB-MRI allows early detection of tumors in variant carriers; and in cancer patients with germline disease-causing TP53 variants, radiotherapy, and conventional genotoxic chemotherapy contribute to the development of subsequent primary tumors. It is critical to perform TP53 testing before the initiation of treatment in order to avoid in carriers, if possible, radiotherapy and genotoxic chemotherapies. In children, the recommendations are to perform clinical examination and abdominal ultrasound (US) every 6 months, annual WB-MRI and brain MRI from the 1st year of life, if the TP53 variant is known to be associated with childhood cancers. In adults, the surveillance should include every year clinical examination, WB-MRI, breast MRI in women from 20 until 65 years and brain MRI until 50 years.

Hanson and colleagues (2020) stated that constitutional pathogenic variants in TP53 are associated with LFS or the more recently described heritable TP53-related cancer syndrome and are associated with increased lifetime risks of a wide spectrum of cancers. Due to the broad tumor spectrum, surveillance for this patient group has been limited. To-date, the only recommendation in the United Kingdom (UK) has been for annual breast MRI in women; however, more recently, a more intensive surveillance protocol including WB-MRI has been recommended by International Expert Groups. To address the gap in surveillance for this patient group in the UK, the UK Cancer Genetics Group facilitated a 1-day consensus meeting to discuss a protocol for the UK. Using a pre-workshop survey followed by structured discussion on the day, these researchers achieved consensus for a UK surveillance protocol for TP53 carriers to be adopted by UK Clinical Genetics services. The key recommendations are for annual WB-MRI and dedicated brain MRI from birth, annual breast MRI from 20 years in women and three-four monthly abdominal US in children along with review in a dedicated clinic.

Concul and co-workers (2020) noted leukemia and tumors of the brain, soft tissues, breasts, adrenal glands, and bone are the most common cancers associated with LFS. Patients with LFS are very susceptible to radiation; thus, the use of WB-MRI is recommended for regular cancer screening. It is important to recognize the common tumors associated with LFS on MRI, and it is also important to be aware of the high rate of false-positive (FP) lesions. The authors concluded that WB-MRI is useful for the detection of cancer in patients who come for regular screening; however, it is associated with pitfalls about which the radiologist must remain aware.

Magnetic Resonance Imaging for Collateral Ligament Injuries of the Lesser Digits' Proximal Interphalangeal Joints

Sahin (2022) noted that collateral ligament injuries of the thumb and lesser digits are simple injuries; however, they may result in disabilities in hand function. In a retrospective study, the author examined the accuracy and cost-effectiveness of MRI in diagnosing proximal interphalangeal (PIP) collateral ligament injuries of lesser digits. This trial was carried out on 18 fingers that had undergone surgery for PIP joint complete collateral ligament injury. Pre-operative MRI results were compared with the intra-operative findings. The data from MRI and direct intra-operative findings were analyzed by the Chi-square test in paired groups. The McNemar test analyzed the accuracy of the MRI test for detecting volar plate injuries. Statistical Packages for Social Sciences (SPSS) version 25 software program was employed for the analysis. In digits other than the thumb, the accuracy of MRI for detecting collateral injuries was 38.89 %, and detection was incorrect in 11 (61.11 %) of 18 patients. There are significant differences between MRI and Intra-operative results ($p < 0.001$). MRI findings for 7 fingers (38.89 %) of the 18 fingers involved were compatible with the surgery results (38.88 %). By comparison, the MRI findings of 11 fingers (61.11 %) were inconsistent with the intra-operative results. A total of 8 patients (44.44 %) were diagnosed pre-operatively with MRI as having volar plate ruptures, 3 (16.67 %) were diagnosed with open surgery, but only 3 of the volar plate diagnosed patients with MRI were verified as ruptures during open surgery (38.0 %). Furthermore, pre-operatively undetected volar plate injuries by MRI ($n = 10$) were detected intra-operatively in 3 cases (30.0 %). Thus, the accuracy of MRI was found not to be statistically significant for the detection of volar plate injuries ($p = 0.727$). The authors concluded that the findings of this study showed that a 1.5-Tesla MRI with a slice thickness of 2 to 3 mm should not be relied on as a decisive tool for diagnosing collateral ligament injuries of the PIP joint of the lesser digits. In addition, MRI was found to be insufficient for diagnosing volar plate injuries that accompanied collateral ligament injuries. These researchers stated that given these findings, one might conclude that MRI is not cost-effective in diagnosing collateral ligament injuries of the lesser digits PIP joint.

The authors stated that the limitations of this study included the retrospective nature, limited number of subjects (n = 18 fingers), and the lack of a control group to compare the results. A second group could be formed in which partial tears are diagnosed and treated conservatively through physical examination and stress radiography, and subjects in this group could undergo MRI, funded by the university's scientific fund, to determine the accuracy with which MRI detects partial tears in this group. The model may be used to construct an effective study plan for future research.

Magnetic Resonance Imaging Lymphangiography for Peripheral Lymphedema

Forte et al (2021) stated that in recent years, magnetic resonance imaging lymphangiography (MRI-L) has emerged as a way to predict if patients are candidates for lymphedema surgery, especially lympho-venous anastomosis (LVA). In a systematic review, these investigators examined the literature on the use of MRI-L for pre-operative planning in lymphedema surgery. They hypothesized that MRI-L could add valuable information to the standard pre-operative evaluation of lymphedema patients. On February 17, 2020, these researchers carried out a systematic review of the PubMed/Medline, Cochrane Clinical Answers, and Embase databases, without timeframe or language limitations, to identify studies on the use of MRI-L for pre-operative planning of lymphedema surgery. They excluded studies that examined other uses of MRI, such as lymphedema diagnosis and treatment evaluation. The primary outcome was the examination capacity to identify lymphatic anatomy and the secondary outcome was the presence of adverse effects. Of 372 potential studies identified with the search, 9 met the eligibility criteria. A total of 334 lymphedema patients were enrolled in these studies. Two studies compared MRI-L findings with those of other standard examinations (indocyanine green lymphography [ICG-L] or lymphoscintigraphy). No adverse effects due to MRI-L were reported. A study reported that MRI-L had higher sensitivity for the detection of lymphatic vessel abnormalities compared with lymphoscintigraphy and a statistically higher chance of successful LVA was observed when the results of MRI-L agreed with those of ICG-L ($p < 0.001$). The authors concluded that MRL could be useful for pre-operative planning in lymphedema surgery. Moreover, these investigators stated that the evidence has been limited; thus, further investigation with larger sample sizes and cost-analysis are needed to justify the addition of this novel approach to current pre-operative protocols.

The authors stated that drawbacks of this systematic review included the limited number of studies (n = 9) and limited ability to generalize findings. Moreover, there was also a possibility of bias in the analysis of information from each article. Nevertheless, these researchers were able to summarize and discuss the relevance of MRI-L for pre-operative planning of lymphedema surgery, which is relevant to the development of a definitive surgical therapy for lymphedema in appropriate patients. Certainly, cost-analysis studies are needed to support the use of MRI-L as the standard of care (SOC) for pre-operative planning before performing lymphedema surgery.

Quartuccio et al (2022) noted that peripheral lymphedema represents a disabling condition affecting the lymphatic system of the limbs resulting from impaired drainage and excessive lymphatic fluid accumulation in the interstitial spaces. Lymphoscintigraphy stands as the imaging modality of the 1st choice to evaluate patients with peripheral lymphedema. Nevertheless, in recent times, MRI techniques have also been employed to examine patients with lymphedema. In a systematic review, these investigators examined the available evidence providing a head-to-head comparison between lymphoscintigraphy and MRI techniques in peripheral lymphedema. They carried out a systematic literature search using the PubMed database and Cochrane Central Register of Controlled Trials (CENTRAL). The eligibility criteria for the studies to be included in the qualitative synthesis included a study cohort or a subset of patients with a clinical diagnosis of peripheral lymphedema (either upper or lower limb); execution of both MRI and lymphoscintigraphy in the same subset of patients. The methodological quality of the studies was assessed by an investigator using the "Quality Assessment of Diagnostic Accuracy Studies" tool, v. 2 (QUADAS-2). A total of 11 studies were included in the quantitative analysis. No meta-analysis was carried out due to the heterogeneous patient samples, the different study objectives of the retrieved literature, and the limited number of available articles. In the diagnosis of upper limb (UL) extremity lymphedema, the sensitivity of MRI techniques appeared superior to that of lymphoscintigraphy. Comparative studies in the LEs are still scarce but suggested that MRI may increase the diagnostic accuracy for lymphedema. The authors concluded that available literature on patients with lymphedema examined with both lymphoscintigraphy and MRI did not allow definite conclusions on the superiority of one imaging technique over the other one. These researchers stated that further studies including well-selected patient samples are needed to compare the accuracy of these imaging modalities. Since MRI techniques appeared to provide complementary findings to lymphoscintigraphy, it would be conceivable to acquire both imaging studies in patients with peripheral lymphedema. Furthermore, these investigators stated that studies examining the clinical impact of adding MRI to the diagnostic work-up are needed.

Magnetic Resonance Imaging for Evaluation of Ankle Instability

de Krom et al (2022) noted that supination-external rotation (SER) ankle fractures account for the majority of ankle fractures and can be divided into stable or unstable fractures, based on the state of the deltoid ligament. In a systematic review and meta-analysis, these investigators examined the available evidence on diagnostic tools for evaluating deltoid ligament integrity in patients with SER-type ankle fractures. They carried out a comprehensive literature search of PubMed and Embase up to December 2020. The outcome measures were sensitivity, specificity and positive predictive value (PPV) and negative predictive value (NPV) of the diagnostic tools. A meta-analysis was carried out to obtain an overview of sensitivity, specificity and area under the curve (AUC). The methodological quality of the studies was examined using Quality Assessment of Diagnostic Accuracy Studies. A total of 12 studies examining tools for deltoid ligament rupture in patients with SER-type ankle fractures were included. The present study found sensitivity (and specificity) ranges of 0.20 to 0.90 (and 0.38 to 0.97) for clinical features, MRI 0.57 to 0.85 (and 0.81 to 1.00), US 1.00 (and 0.89 to 1.00), malleolar medial fleck sign (MMFS) 0.25 (and 0.99), conventional ankle mortise radiography 0.33 to 0.57 (and 0.60 to 0.94), gravity stress radiography 0.71 to 1.00 (and 0.72 to 0.88) and manual stress ankle radiography 0.65 to 1.00 (and 0.00 to 0.77). The largest AUC was found for US, followed by MMFS, gravity stress radiography and MRI. The authors concluded that US and gravity stress radiography appeared to be the most accurate diagnostic tools to evaluate deltoid ligament integrity. To strengthen this conclusion, future research should focus on better overall methodology, especially regarding the use of an identical reference test. Nevertheless, this review was of high value to close the knowledge gap regarding which presently available diagnostic tool is to be preferred to evaluate deltoid ligament integrity in patients with SER-type ankle fractures.

The authors stated that this systematic review had several drawbacks. First, the most important weakness of this review was the fact that various reference tests were used since no consensus exists regarding the gold standard. Second, the studies used different cut-off points to define a positive result. Third, these researchers did not examine the risk of publication bias of the studies included; however, the assumption was that the risk of publication bias was relatively low since the studies included also showed negative results regarding the ability for diagnosing deltoid ligament rupture. Fourth, all studies examined the diagnostic accuracy of single diagnostic methods, but not all methods were studied simultaneously in 1study and in the same study population.

Magnetic Resonance Imaging of the Knee for Chronological Age Estimation

Muller et al (2023) noted that radiographs of the hand and teeth are often used for medical age assessment, as skeletal and dental maturation correlates with chronological age. These methods have been criticized for their lack of precision, and MRI of the knee has been proposed as a more accurate method. In a systematic review, these investigators examined the available evidence for medical age estimation based on skeletal maturation as evaluated by MRI of the knee. They carried out a systematic review that included studies published before April 2021 on living individuals between age of 8 and 30 years, with presumptively healthy knees for whom the ossification stages had been examined using MRI. The correlation between "mature knee" and chronological age as well as the risk of misclassifying a child as an adult and vice versa was calculated. These researchers found a considerable heterogeneity in the published studies in terms of study population, MRI protocols, as well as grading systems used. There was a wide variation in the correlation between maturation stage and chronological age. The authors concluded that data from published literature is deemed too heterogenous to support the use of MRI of the knee for chronological age determination. Furthermore, it was not possible to evaluate the sensitivity, specificity, negative predictive value (NPV), or positive predictive value (PPV) for the ability of MRI to determine whether a person is over or under 18 years of age.

The authors stated that the key drawback of this systematic review was that relatively few studies were identified. Larger and more comparable studies may be able to show a stronger trend in terms of the proportion of individuals per age group deemed to have a mature knee on MRI. Still, these investigators will not be able to predict the risk of misclassification in individuals with an unknown age due to the statistical paradox described. Forensic age estimations can have huge legal implications for the individual being evaluated, as well as for countries and authorities who use such methods. Thus, the methods used should be reliable and reproducible and the statistical calculations of probability must exactly fit the population to which it is applied.

Matijas et al (2023) stated that the knee is an anatomical structure that can provide a great deal of data for research on age estimation. The se researchers examined and used a method for semi-automatic measurements of the area under the growth plate closure of the femur distal epiphysis and the growth plate closure itself on the 2D coronary slices using T2 weighted images (T2WI) generated on MRI devices of different technical and technological characteristics. After the semi-automatic segmentation of the femur distal epiphysis under the growth plate closure and the growth plate closure itself, the areas of the measured closures were calculated using MATLAB version: 9.12. (R2022a) for each individual coronal slice . The area ratio index (ARI) was calculated as the ratio between the area under the growth plate closure of the femur distal epiphysis and the growth plate closure itself. The study sample consisted of 27 female and 23 male Caucasian subjects aged 10 to 26 years. A total of 339 T2WI images were used for ARI calculations. There was a positive correlation between chronological age and the average ARI measured by 3 independent observers ($r = 0.8280$, $p < 0.001$). The authors concluded that the findings of this study showed that using MRI devices with different technical and technological characteristics for measurements of the area under the growth plate closure of the femur distal epiphysis and the growth plate closure itself did not influence the results of the ARI used for age estimation; thus, such methods may guide future studies, help researchers decide on a preferred approach for specific cases, and contribute to multi-factorial age estimation better than based on a single anatomical site.

The authors stated that the drawbacks of this study included the small sample size ($n = 50$ subjects). Although a total of 339 images were analyzed, further investigations are needed to collect more data, which may improve the precision of this method. Furthermore, considering the retrospective design of the study, there was a lack of information regarding sports activities, socio-economic status, as well as possible diseases, especially endocrine diseases. This method was built upon data from healthy youth and young adult subjects, and the effects of disorders that could affect growth were not examined.

Functional MRI for Evaluation of Upper Extremity Impairment After Stroke

Li et al (2024) noted that changes in cortical excitability after stroke are closely associated with motor function recovery. These researchers examined the motor network re-organization mechanisms corresponding to the different clinical outcomes of upper extremity (UE) motor impairment in patients with subacute stroke. Motor function was examined before rehabilitation (pre), after rehabilitation (post), and at the 1-year follow-up (follow-up) using the Fugl-Meyer assessment UE scale. Additionally, resting-state functional MRI (fMRI) data were collected in both pre- and post-conditions. A total of 20 patients with stroke were categorized into good and poor outcome groups based on motor impairments at the 1-year follow-up. Functional connections between motor-related regions of interest and the rest of the brain were subsequently calculated. Lastly, the correlation between motor network re-organization and behavioral improvement at the 1-year follow-up was examined. The good outcome group exhibited a positive pre-condition motor function and continuous improvement, whereas the poor outcome group revealed a weak pre-condition motor function and insignificant improvement. Contra-lesional hemisphere-related connections were found to be higher in the good outcome group pre-conditioning, with both groups showing minimal change post-conditioning, while no relationship with motor impairment was observed. Long inter-hemispheric connections were decreased and increased in the good and poor outcome groups, respectively, and were negatively correlated with motor impairment. The authors concluded that different motor network re-organizations during the subacute phase could influence the varying motor outcomes in the affected UE after stroke. Moreover, these researchers stated that these findings may serve as the theoretical basis for future neuro-modulatory research.

Quantitative MRI for Evaluation of Muscle Microtrauma in the Lower Extremity

Cheng and Li (2024) examined the available evidence and discussed recent advances in quantitative MRI techniques for the evaluation of lower extremity (LE) muscle microtrauma after a marathon. Single-modality quantitative MRI techniques include T2 mapping to examine the dynamics of muscle inflammatory edema and variability at the site of injury, diffusion tensor imaging (DTI) to detect sub-clinical changes in muscle injury, intravoxel incoherent motion (IVIM) imaging to provide simultaneous information on perfusion and diffusion in muscle tissue without the need for intravenous (IV) contrast, and magnetic resonance spectroscopy (MRS) to non-invasively detect intra-myocellular lipid (IMCL) content in muscle before and after marathon exercise to explain the use of fatty acids as an energy source in skeletal muscle during long-distance running. Chemical exchange saturation transfer (CEST) is especially suitable for detecting changes in free creatine, pH values, and lactate concentrations in muscles before and after exercise, providing a more detailed picture of muscle physiology and chemistry. These metabolic MRI methods enhance the understanding of biochemical alterations occurring in muscles pre- and post-exercise. Multi-modal techniques combine different modalities to provide a comprehensive evaluation of muscle structural and functional changes. The authors concluded that these advanced techniques aim to better evaluate muscle microtrauma and guide clinical treatment. Moreover, these researchers stated that prospective, longitudinal studies with large sample sizes are needed to determine standardized processes for these technologies.

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